ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

Rabaa Farrag

01120486569

23 YEARS OF EXPERIENCE

Reach a 700+ in ACT, Digital SAT & EST Math

✓ FIRST SESSION IS FREE!

Don't watch others hit 700+ while you wait.

ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

# ACT2 Exam 5

## Full Practice Test • Test #5

Polygons, circles, trigonometry, factoring & radicals. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

| 50 | MCQ | 60 min |
|---|---|---|
| QUESTIONS | MULTIPLE CHOICE | TIME LIMIT |

NAME _____ DATE _____ SCORE _____ /50

### How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

**Eng. Rabaa Farrag**

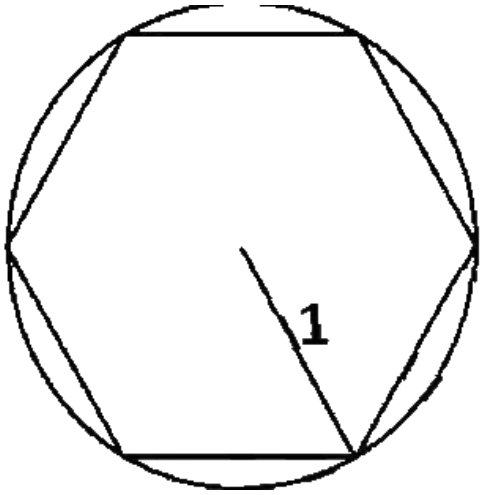
ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #5

# Practice Questions

50 QUESTIONS

**1** Find the perimeter of the following hexagon.

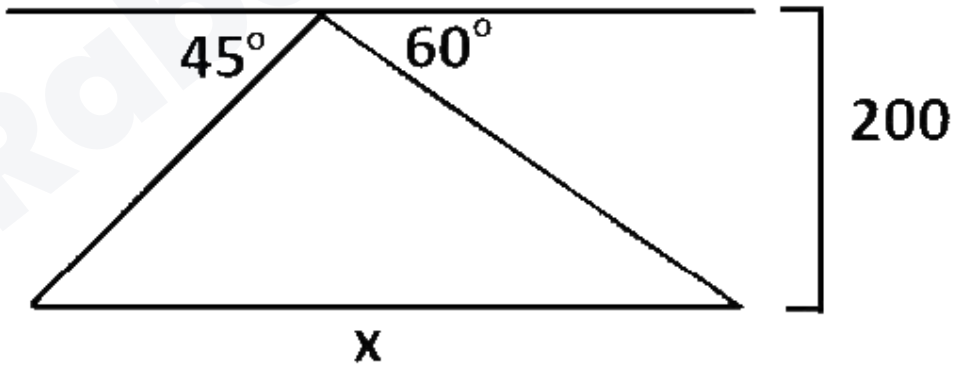



(A) 3
(B) 5
(C) 6
(D) 12

**2** If the area of a triangle is $A = \frac{1}{2}bh$, write $h$ in terms of $A$ and $b$.

(A) $h = \frac{b}{2A}$
(B) $h = \frac{2A}{b}$
(C) $h = \frac{A}{2b}$
(D) $h = -\frac{2A}{b}$

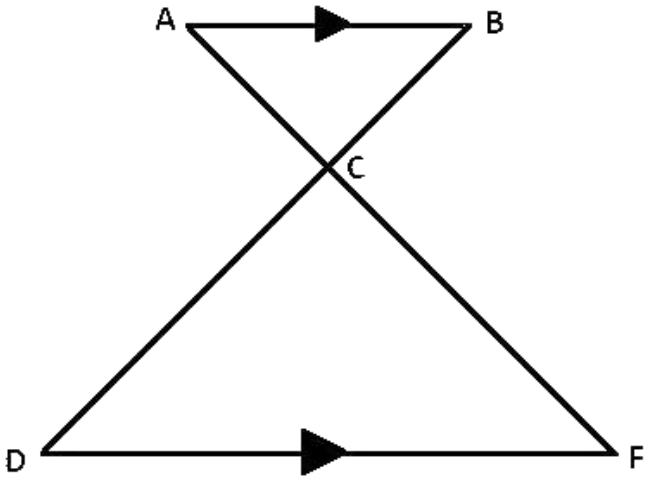
**3** Find $x$.




(A) 314.25
(B) 315
(C) 315.5
(D) 316

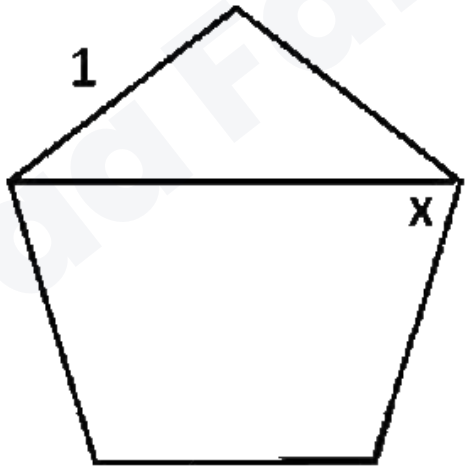
Rabaa Farrag

**4** Are triangles $ABC$ and $DFC$ similar?




(A) Yes, similar by SAS
(B) No, not enough information
(C) Yes, similar by AA
(D) Similar only if $AE \parallel BD$

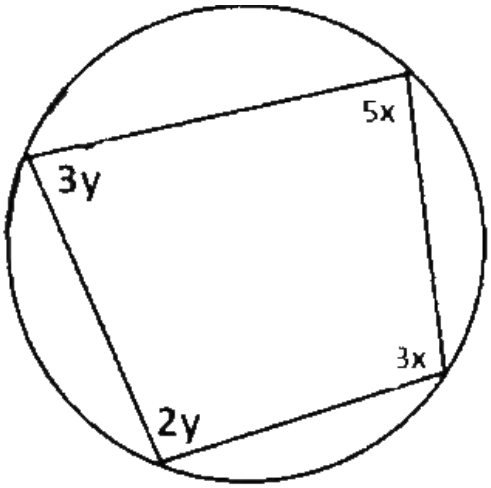
**5** Find $x$ in this regular polygon.




(A) 108
(B) 36
(C) 72
(D) 24

Rabaa Farrag

**6** What are $x$ and $y$?

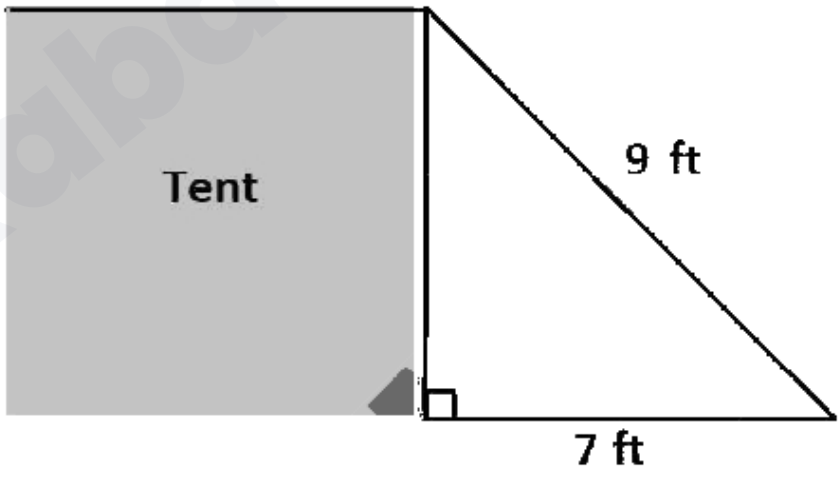



(A) $x = 20, y = 40$
(B) $x = 40, y = 20$
(C) $x = 30, y = 60$
(D) $x = -180, y = 360$

**7** If $\frac{a^n \cdot a^m}{a^k} = 1$, which statement is true?

(A) $m + n = k$
(B) $\frac{n+m}{k} = 1$
(C) $\frac{nm}{k} = 0$
(D) $\frac{m}{k} = n$

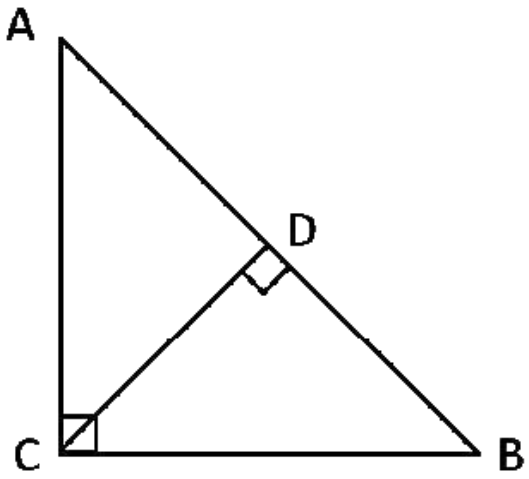
**8** A camper fastened a rope to the top of a tent and tied the other end to a stake in the ground, 7 ft away. The length of the rope from the tent to the stake is 9 ft. What is the tangent of the angle that the rope makes with the level ground?




(A) $\frac{4\sqrt{2}}{7}$
(B) $\frac{4\sqrt{2}}{9}$
(C) $\frac{7}{9}$
(D) $\frac{7\sqrt{2}}{8}$

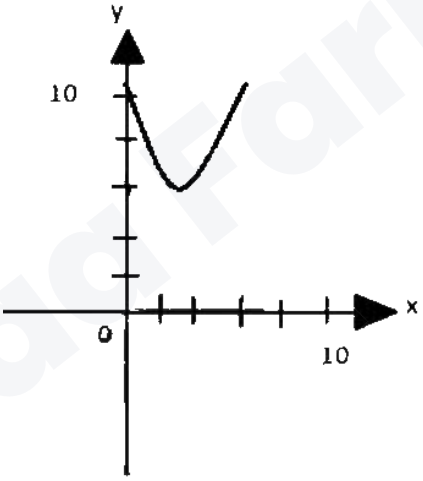
Rabaa Farrag

**9** In right $\triangle ACB$, $AB = 4$, $\overline{CA} \cong \overline{CB}$, and $\overline{CD} \perp \overline{AB}$. What is $CD$?




(A) $1$
(B) $2$
(C) $2\sqrt{2}$
(D) $4$

**10** Which of the following is the equation of the parabola?

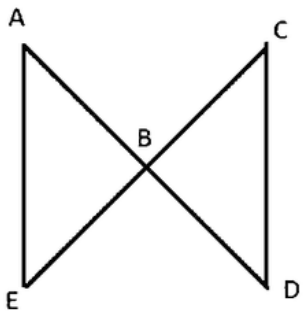



(A) $y = 2(x-3)^2 + 2$
(B) $y = 2(x+3)^2 + 2$
(C) $y = \frac{1}{2}(x-3)^2 + 6$
(D) $y = \frac{1}{2}(x-3)^2 + 2$

Rabaa Farrag

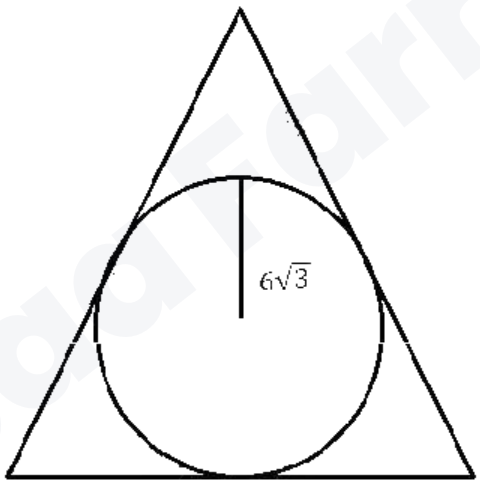
**11** Given $\overline{AD}$ and $\overline{CE}$ bisect each other. Prove: $\triangle ABE \cong \triangle DBC$. What is the missing reason in step 4?

| Statement | Reason |
|---|---|
| 1- Given $\overline{AD}$ and $\overline{CE}$ bisect each other | Given |
| 2- $\overline{AB} \cong \overline{DB}$ $\overline{EB} \cong \overline{BC}$ | Definition of a segment bisector |
| 3- $\angle ABE \cong \angle DBC$ | Vertical angels Are congruent |
| 4- $\Delta ABE = \Delta DBC$ | ?? |




(A) SAS
(B) SSS
(C) AAS
(D) ASA

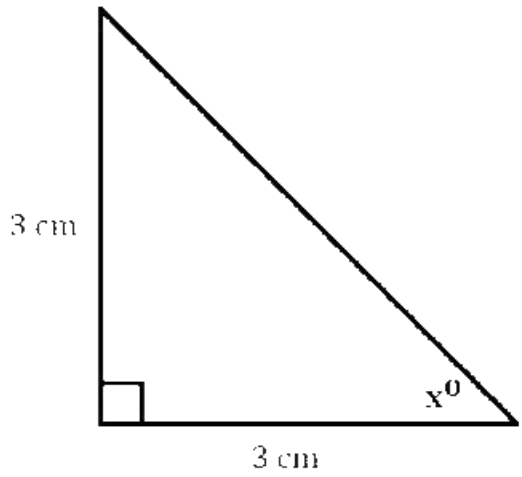
**12** A gardener designed a circular flower bed surrounded by bushes that approximate an equilateral triangle. He planted bushes at each vertex of the triangle and at $6$ ft intervals along its boundary. Each side of the triangle is tangent to the circle, which has a radius of $6\sqrt{3}$ ft. How many bushes did he use?




(A) 6
(B) 18
(C) 36
(D) 54

Rabaa Farrag

**13** In the triangle in the figure below, what is the value of $x$?




(A) 30
(B) 40
(C) 45
(D) 60

**14** What is the exterior angle of a regular pentagon?

(A) 72
(B) 90
(C) 108
(D) 120

**15** During a 5-hour snowstorm, the equation $y = \frac{3}{4}x + 2$ models the depth of snow on the ground $y$ after $x$ hours. What is the slope?

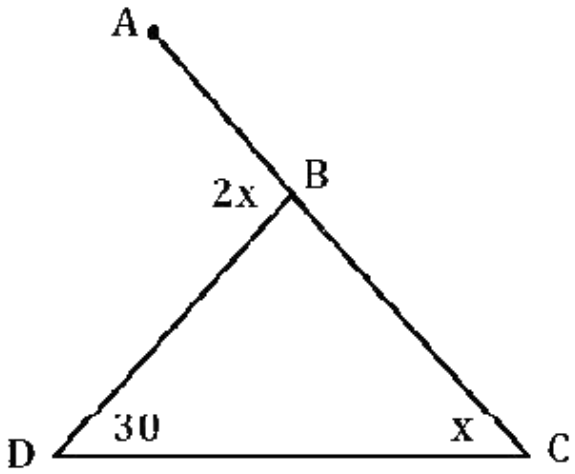
(A) 2
(B) 3
(C) 4
(D) $\frac{3}{4}$

**16** Simplify $3(4 - x^2) + 10$.

(A) $3x^2 - 22$
(B) $3x^2 + 22$
(C) $-3x^2 - 22$
(D) $-3x^2 + 22$

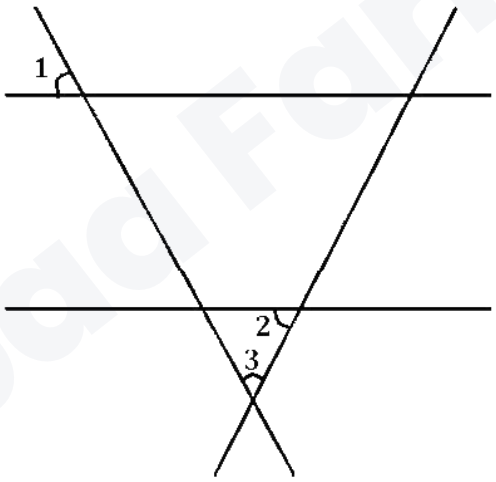
Rabaa Farrag

**17** In $\triangle ABC$, $m\angle BDC = 30°$. If the measure of exterior angle $\angle ABD$ is twice the measure of the interior angle $\angle DCB$, what is $m\angle ABD$?




(A) $75°$
(B) $60°$
(C) $90°$
(D) $120°$

**18** A map of $4$ roads intersect. $m\angle 1 = 71°$ and $m\angle 2 = 43°$. Find $m\angle 3$.




(A) $71°$
(B) $43°$
(C) $66°$
(D) $76°$

**19** Simplify $2(a - b) - a(3 - b)$.

(A) $a + 2b - ab$
(B) $a - 2b + ab$
(C) $-a - 2b + ab$
(D) $-a + 2b - ab$

Rabaa Farrag

**20** Factor $4x^2 - 12xy + 9y^2$.

(A) $(3x - 2y)(3x + 2y)$
(B) $(2x - 3y)(2x + 3y)$
(C) $(2x - 3y)(2x - 3y)$
(D) $(2x + 3y)(2x + 3y)$

**21** Solve $2|3x - 8| - 3 = 11$.

(A) $x = \frac{1}{5}$ and $x = 3$
(B) $x = \frac{1}{3}$ and $x = 5$
(C) $x = 5$ and $x = 3$
(D) $x = 5$ and $x = -3$

**22** Given $x \neq 0$, simplify $\frac{2}{3x} + \frac{4}{3x}$.

(A) $\frac{x}{2}$
(B) $\frac{2}{x}$
(C) $x$
(D) $x^2$

**23** If $f(x) = x^2 - 3x - 1$, find $f(-2)$.

(A) $8$
(B) $9$
(C) $-9$
(D) $-8$

**24** What is the square root of $25 \cdot 3^{16}$?

(A) $328.05$
(B) $32805$
(C) $33805$
(D) $34800$

**25** Simplify $(4x^2 - 5x + 6)(8x - 3)$.

(A) $32x^3 - 52x^2 + 63x - 18$
(B) $16x^3 - 26x^2 + 63x - 9$
(C) $-32x^3 + 52x^2 + 63x - 18$
(D) $-32x^3 - 52x^2 - 63x - 18$

**26** Simplify $(2x^2y)^3$.

(A) $4x^8y^6$
(B) $16x^6y^3$
(C) $8x^6y^3$
(D) $48y^6$

Rabaa Farrag

**27** Find the difference $(x^2 + 4x + 4) - (5x^2 - x - 2)$.

A $-4x^2 + 5x + 6$
B $6x^2 + 3x + 2$
C $4x^2 - 5x - 6$
D $-6x^2 - 3x - 2$

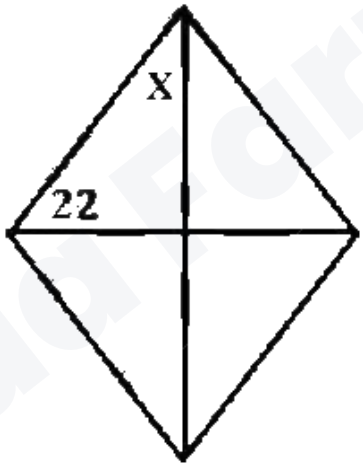
**28** Simplify $\left(\sqrt{16x^6}\right)\left(\sqrt{9x^4}\right)$.

A $12x^5$
B $16x^6$
C $-12x^5$
D $12x^6$

**29** Solve $3x - 6 < 18$.

A $x < 8$
B $x > 8$
C $x \leq 8$
D $x \geq 8$

**30** Find $x$ in the rhombus below.




A 60
B 75
C 90
D 68

**31** Find the equation of the line whose slope is $\frac{2}{3}$ and passing through the point $(-2, 5)$.

A $y = \frac{2}{3}x + \frac{19}{3}$
B $y = -\frac{2}{3}x + \frac{19}{3}$
C $y = \frac{3}{2}x + \frac{19}{3}$
D $y = \frac{2}{3}x - \frac{19}{3}$

Rabaa Farrag

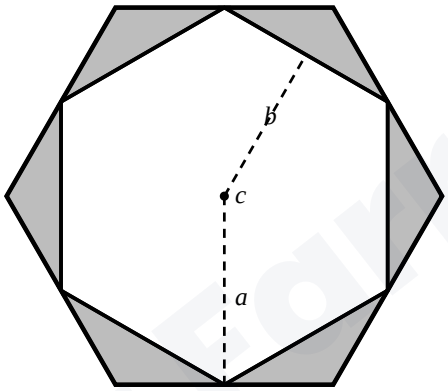
**32** Factor $81m^2 - 4b^2$.

(A) $(9m - 2b)(9m + 2b)$
(B) $(9m + 2b)(9m + 2b)$
(C) $(9m - 2b)(9m - 2b)$
(D) $9(m - 2b)(m + 2b)$

**33** A circle is inscribed in a square with side length $20$. Find the circle's circumference.

(A) $c = 20$
(B) $c = 10\pi$
(C) $c = \frac{\pi}{20}$
(D) $c = 20\pi$

**34** Two regular hexagons with center $C$ and apothems $a$ and $b$ are shown in the figure below. Each vertex of the smaller hexagon is a midpoint on a side of the larger hexagon. If $a = 12\sqrt{3}$ cm and $b = 18$ cm, what is the total area of the shaded regions in $\text{cm}^2$?




(A) $36\sqrt{3}$
(B) $216\sqrt{3}$
(C) $648\sqrt{3}$
(D) $864\sqrt{3}$

**35** Mohammed bought $5$ pizzas, each at a price $p$, $3$ cans of soda each at price $s$, and $1$ bottle of juice at price $j$. Write the expression that represents the total amount he paid for the pizza, soda and juice.

(A) $5p + 3s + j$
(B) $3p + 3s + j$
(C) $5p + s + 3j$
(D) $p + 5j + 3s$

**36** If $x^2 - 6 = 0$, find the zeros.

(A) $-\sqrt{6}$
(B) $\sqrt{6}$
(C) $3$
(D) $\pm\sqrt{6}$

Rabaa Farrag

**37** According to the following table, if Jacob's weight is $160$ lb, what is his width?

| Width | Weight |
|---|---|
| 10 | 100 |
| 11 | 110 |
| 12.3 | 123 |
| 14.9 | 149 |

A) $15$
B) $14$
C) $16$
D) $17$

**38** The zeros of a quadratic function are $x = -2$ and $x = 3$. What is the equation in factored (product) form?

A) $(x+3)(x-2)$
B) $(x-3)(x-2)$
C) $(x+2)(x-3)$
D) $(x+2)(x+3)$

**39** If the circumference of a circle is $20$, find the radius.

A) $R = 10\pi$
B) $R = \frac{\pi}{10}$
C) $R = \frac{10}{\pi}$
D) $R = \pi$

**40** Simplify $\frac{\sqrt{8x^3}}{\sqrt{27x}}$.

A) $\frac{\sqrt{6}}{9}x$
B) $\frac{2\sqrt{6}}{9}x$
C) $\frac{4\sqrt{6}}{9}x$
D) $\frac{9}{2\sqrt{6}}x$

**41** Factor $18x^2 - 24x + 8$.

A) $2(3x-2)(3x-2)$
B) $6(x-2)(x-2)$
C) $2(3x+2)(3x+2)$
D) $(3x-2)(3x-2)$

**42** If $\tan x = \frac{5}{12}$, find $\sin x + \cos x$ (for acute $x$).

A) $\frac{13}{17}$
B) $\frac{2}{12}$
C) $\frac{17}{13}$
D) $1$

Rabaa Farrag

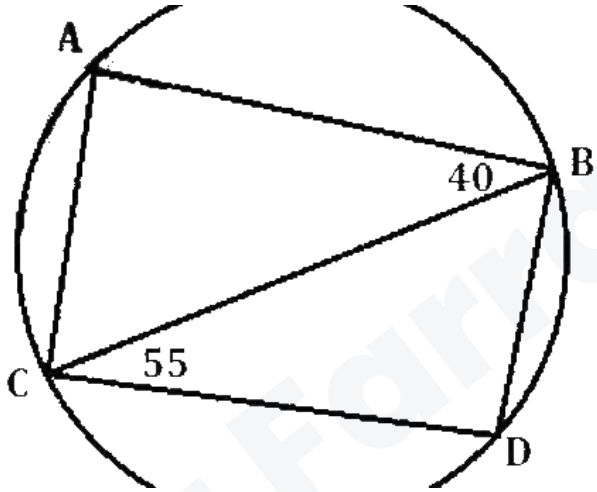
**43** Solve $-x + 3 \geq 6$.

(A) $x \geq 3$
(B) $x \leq 3$
(C) $x \leq -3$
(D) $x \geq -3$

**44** Simplify $\sqrt{\frac{36}{48}}$.

(A) $\sqrt{3}$
(B) $2$
(C) $\frac{2}{\sqrt{3}}$
(D) $\frac{\sqrt{3}}{2}$

**45** Find the sum of the measures of $\angle ACB$ and $\angle CBD$.




(A) $75$
(B) $85$
(C) $95$
(D) $105$

**46** $y = 0.34x - 3$ and $x$ represents time in minutes. Find $y$ if the time was $3$ hours and $20$ min.

(A) $60$
(B) $63$
(C) $65$
(D) $80$

**47** Simplify $-2x(2x - 1)(x + 2)$.

(A) $-4x^3 + 10x^2 + 4x$
(B) $-4x^3 + 5x^2 + 2x$
(C) $-4x^3 - 10x^2 + 4x$
(D) $-4x^3 - 6x^2 + 4x$

**48** If the mean of $3$ marks is $80$ and the mean of another $4$ marks is $70$, find the mean of all the marks.

(A) $\frac{240}{7}$
(B) $\frac{520}{7}$
(C) $\frac{280}{7}$
(D) $\frac{280}{4}$

Rabaa Farrag

**49** $f(x) = x^2 + 3$ and $g(x) = 5x - 9$. Find $f(x) - g(x)$.

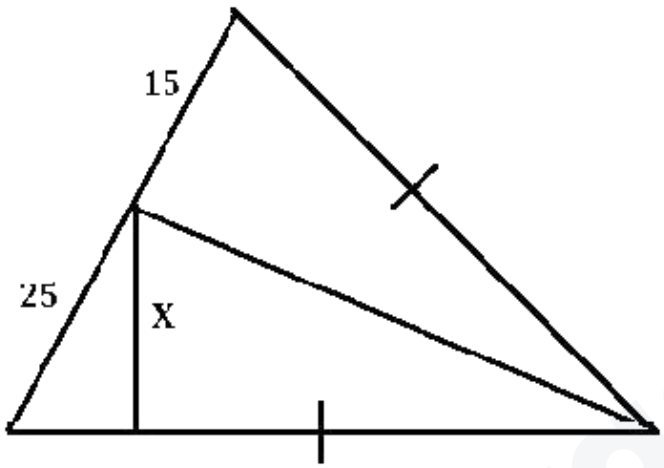
A $-x^2 + 5x - 12$

B $x^2 - 5x + 12$

C $-2x^2 + 10x - 24$

D $x^2 - 5x$

**50** Find $x$.




A 15

B 25

C 12.5

D 30

Rabaa Farrag

TEST #5 • CHECK YOUR WORK

# Answer Key

50 ANSWERS

| Q | ANSWER | Q | ANSWER |
|---|---|---|---|
| 1 | C | 26 | C |
| 2 | B | 27 | A |
| 3 | C | 28 | A |
| 4 | C | 29 | A |
| 5 | C | 30 | D |
| 6 | A | 31 | A |
| 7 | A | 32 | A |
| 8 | A | 33 | D |
| 9 | B | 34 | B |
| 10 | C | 35 | A |
| 11 | A | 36 | D |
| 12 | B | 37 | C |
| 13 | C | 38 | C |
| 14 | A | 39 | C |
| 15 | D | 40 | B |
| 16 | D | 41 | A |
| 17 | B | 42 | C |
| 18 | C | 43 | C |
| 19 | C | 44 | D |
| 20 | C | 45 | B |
| 21 | B | 46 | C |
| 22 | B | 47 | D |
| 23 | B | 48 | B |
| 24 | B | 49 | B |
| 25 | A | 50 | A |

Rabaa Farrag

TEST #5 • STEP-BY-STEP

# Detailed Solutions

50 SOLUTIONS

**1** **Answer: C** 6

**Step 1:** A regular hexagon inscribed in a circle splits into 6 equilateral triangles from the center.
**Step 2:** So each side equals the radius: side $= 1$.
**Step 3:** Perimeter $= 6 \times 1 = 6$.

**2** **Answer: B** $h = \frac{2A}{b}$

**Step 1:** Multiply both sides by 2: $2A = bh$.
**Step 2:** Divide by $b$: $h = \frac{2A}{b}$.

**3** **Answer: C** 315.5

**Step 1:** Drop the height 200 from the top vertex; it splits the base into two parts.
**Step 2:** Left part $(45°)$: $\frac{200}{\tan 45°} = 200$. Right part $(60°)$: $\frac{200}{\tan 60°} = \frac{200}{\sqrt{3}} \approx 115.5$.
**Step 3:** $x \approx 200 + 115.5 = 315.5$.

**4** **Answer: C** Yes, similar by AA

**Step 1:** The arrows show $\overline{AB} \parallel \overline{DF}$.
**Step 2:** Alternate interior angles are congruent: $\angle A \cong \angle F$ and $\angle B \cong \angle D$; also the vertical angles at $C$ are equal.
**Step 3:** Two pairs of equal angles: the triangles are similar by AA.

**5** **Answer: C** 72

**Step 1:** Each interior angle of a regular pentagon is $\frac{(5-2)\cdot 180°}{5} = 108°$.
**Step 2:** The diagonal cuts off the top isosceles triangle with apex $108°$: base angles $= \frac{180-108}{2} = 36°$.
**Step 3:** $x = 108° - 36° = 72°$.

**6** **Answer: A** $x = 20, y = 40$

**Step 1:** Opposite angles of a quadrilateral inscribed in a circle add to $180°$.
**Step 2:** $3y + 3x = 180 \Rightarrow x + y = 60$, and $5x + 2y = 180$.
**Step 3:** Substitute $y = 60 - x$: $5x + 120 - 2x = 180 \Rightarrow x = 20, y = 40$.

**7** **Answer: A** $m + n = k$

**Step 1:** Combine the exponents: $\frac{a^n \cdot a^m}{a^k} = a^{n+m-k}$.
**Step 2:** $a^{n+m-k} = 1 = a^0 \Rightarrow n + m - k = 0$.
**Step 3:** $m + n = k$.

Rabaa Farrag

**8** **Answer: A** $\frac{4\sqrt{2}}{7}$

**Step 1:** The tent, the ground and the rope form a right triangle: hypotenuse $9$ ft, base $7$ ft.
**Step 2:** Tent height $= \sqrt{9^2 - 7^2} = \sqrt{32} = 4\sqrt{2}$ ft.
**Step 3:** $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{4\sqrt{2}}{7}$.

**9** **Answer: B** $2$

**Step 1:** $CA = CB$ with a right angle at $C$: an isosceles right triangle with hypotenuse $AB = 4$.
**Step 2:** The altitude $\overline{CD}$ to the hypotenuse is also a median, so $D$ is the midpoint and $AD = DB = 2$.
**Step 3:** $\triangle ADC$ is also isosceles right, so $CD = AD = 2$.

**10** **Answer: C** $y = \frac{1}{2}(x-3)^2 + 6$

**Step 1:** The lowest point (vertex) of the parabola is at $(3, 6)$, so $y = a(x-3)^2 + 6$.
**Step 2:** Only C has the vertex $(3, 6)$: $y = \frac{1}{2}(x-3)^2 + 6$.

**11** **Answer: A** SAS

**Step 1:** Step 2 gives two pairs of congruent sides: $AB \cong DB$ and $EB \cong BC$.
**Step 2:** Step 3 gives the angle between them: $\angle ABE \cong \angle DBC$ (vertical angles).
**Step 3:** Side–Angle–Side: SAS.

**12** **Answer: B** $18$

**Step 1:** For an equilateral triangle, the inradius is $r = \frac{s}{2\sqrt{3}}$, so $s = 2\sqrt{3}\,r = 2\sqrt{3} \cdot 6\sqrt{3} = 36$ ft.
**Step 2:** Perimeter $= 3 \times 36 = 108$ ft.
**Step 3:** Bushes every $6$ ft around a closed loop (vertices included): $\frac{108}{6} = 18$ bushes.

**13** **Answer: C** $45$

**Step 1:** Both legs are $3$ cm, so the triangle is an isosceles right triangle.
**Step 2:** Its two acute angles are equal and add to $90°$.
**Step 3:** $x = 45°$.

**14** **Answer: A** $72$

**Step 1:** The exterior angles of any convex polygon add to $360°$.
**Step 2:** A regular pentagon has $5$ equal exterior angles: $\frac{360°}{5} = 72°$.

Rabaa Farrag

**15** **Answer: D** $\frac{3}{4}$

**Step 1:** In $y = mx + b$, $m$ is the slope.
**Step 2:** $y = \frac{3}{4}x + 2 \Rightarrow$ slope $= \frac{3}{4}$ (the snow deepens $\frac{3}{4}$ unit per hour).

**16** **Answer: D** $-3x^2 + 22$

**Step 1:** Distribute: $3(4 - x^2) = 12 - 3x^2$.
**Step 2:** $12 - 3x^2 + 10 = -3x^2 + 22$.

**17** **Answer: B** $60°$

**Step 1:** Exterior angle theorem: the exterior angle equals the sum of the two remote interior angles.
**Step 2:** $2x = 30 + x \Rightarrow x = 30°$.
**Step 3:** $m\angle ABD = 2x = 60°$.

**18** **Answer: C** $66°$

**Step 1:** The two horizontal roads are parallel, so the angles at the lower crossing match angles $1$ and $2$: $71°$ and $43°$ (corresponding angles).
**Step 2:** These two angles and $\angle 3$ are the angles of the bottom triangle.
**Step 3:** $m\angle 3 = 180° - 71° - 43° = 66°$.

**19** **Answer: C** $-a - 2b + ab$

**Step 1:** Distribute: $2a - 2b - 3a + ab$.
**Step 2:** Combine: $-a - 2b + ab$.

**20** **Answer: C** $(2x - 3y)(2x - 3y)$

**Step 1:** $4x^2 = (2x)^2$, $9y^2 = (3y)^2$, $12xy = 2(2x)(3y)$.
**Step 2:** Perfect square trinomial: $(2x - 3y)(2x - 3y)$.

**21** **Answer: B** $x = \frac{1}{3}$ and $x = 5$

**Step 1:** Isolate: $2|3x - 8| = 14 \Rightarrow |3x - 8| = 7$.
**Step 2:** $3x - 8 = 7 \Rightarrow x = 5$; $3x - 8 = -7 \Rightarrow x = \frac{1}{3}$.
**Step 3:** Solutions: $x = \frac{1}{3}$ and $x = 5$.

**22** **Answer: B** $\frac{2}{x}$

**Step 1:** Same denominator: add the numerators: $\frac{2+4}{3x} = \frac{6}{3x}$.
**Step 2:** Simplify: $\frac{2}{x}$.

Rabaa Farrag

**23** **Answer: B** 9

**Step 1:** $f(-2) = (-2)^2 - 3(-2) - 1$.
**Step 2:** $= 4 + 6 - 1 = 9$.

**24** **Answer: B** 32805

**Step 1:** $\sqrt{25 \cdot 3^{16}} = \sqrt{25} \cdot \sqrt{3^{16}} = 5 \cdot 3^8$.
**Step 2:** $3^8 = 6561$, so the value is $5 \times 6561 = 32805$.

**25** **Answer: A** $32x^3 - 52x^2 + 63x - 18$

**Step 1:** Multiply each term by $8x$ and by $-3$: $32x^3 - 40x^2 + 48x$ and $-12x^2 + 15x - 18$.
**Step 2:** Combine like terms: $32x^3 - 52x^2 + 63x - 18$.

**26** **Answer: C** $8x^6y^3$

**Step 1:** Raise each factor to the 3rd power: $2^3(x^2)^3y^3$.
**Step 2:** $= 8x^6y^3$.

**27** **Answer: A** $-4x^2 + 5x + 6$

**Step 1:** Distribute the minus sign: $x^2 + 4x + 4 - 5x^2 + x + 2$.
**Step 2:** Combine: $-4x^2 + 5x + 6$.

**28** **Answer: A** $12x^5$

**Step 1:** $\sqrt{16x^6} = 4x^3$ and $\sqrt{9x^4} = 3x^2$.
**Step 2:** Product: $4x^3 \cdot 3x^2 = 12x^5$.

**29** **Answer: A** $x < 8$

**Step 1:** Add 6: $3x < 24$.
**Step 2:** Divide by 3: $x < 8$.

**30** **Answer: D** 68

**Step 1:** The diagonals of a rhombus are perpendicular, so they meet at $90°$.
**Step 2:** In the right triangle at the top, the angles are $22°$, $90°$ and $x$.
**Step 3:** $x = 180° - 90° - 22° = 68°$.

Rabaa Farrag

**31** **Answer: A** $y = \frac{2}{3}x + \frac{19}{3}$

**Step 1:** Point-slope form: $y - 5 = \frac{2}{3}(x + 2)$.
**Step 2:** $y = \frac{2}{3}x + \frac{4}{3} + 5 = \frac{2}{3}x + \frac{19}{3}$.

**32** **Answer: A** $(9m - 2b)(9m + 2b)$

**Step 1:** $81m^2 = (9m)^2$ and $4b^2 = (2b)^2$: a difference of squares.
**Step 2:** $(9m - 2b)(9m + 2b)$.

**33** **Answer: D** $c = 20\pi$

**Step 1:** A circle inscribed in a square touches all four sides, so its diameter equals the side: $d = 20$.
**Step 2:** $C = \pi d = 20\pi$.

**34** **Answer: B** $216\sqrt{3}$

**Step 1:** Large hexagon: apothem $a = 12\sqrt{3}$, so its side is $\frac{2a}{\sqrt{3}} = 24$ and its area is $\frac{3\sqrt{3}}{2}(24)^2 = 864\sqrt{3}$.
**Step 2:** Small hexagon: apothem $b = 18$, so its side is $\frac{2(18)}{\sqrt{3}} = 12\sqrt{3}$ and its area is $\frac{3\sqrt{3}}{2}(12\sqrt{3})^2 = 648\sqrt{3}$.
**Step 3:** Shaded area $= 864\sqrt{3} - 648\sqrt{3} = 216\sqrt{3}\ \text{cm}^2$.

**35** **Answer: A** $5p + 3s + j$

**Step 1:** Pizza: $5$ at $p$ each $= 5p$. Soda: $3$ at $s$ each $= 3s$. Juice: $1$ at $j = j$.
**Step 2:** Total $= 5p + 3s + j$.

**36** **Answer: D** $\pm\sqrt{6}$

**Step 1:** $x^2 = 6$.
**Step 2:** $x = \sqrt{6}$ or $x = -\sqrt{6}$, i.e. $\pm\sqrt{6}$.

**37** **Answer: C** $16$

**Step 1:** In every row, weight $= 10\times$ width ($100 = 10 \cdot 10$, $123 = 10 \cdot 12.3$, …).
**Step 2:** Width $= \frac{160}{10} = 16$.

**38** **Answer: C** $(x + 2)(x - 3)$

**Step 1:** A zero at $x = r$ gives the factor $(x - r)$.
**Step 2:** Zeros $-2$ and $3$: $(x + 2)(x - 3)$.

Rabaa Farrag

**39** **Answer: C** $R = \frac{10}{\pi}$

**Step 1:** $C = 2\pi R = 20$.
**Step 2:** $R = \frac{20}{2\pi} = \frac{10}{\pi}$.

**40** **Answer: B** $\frac{2\sqrt{6}}{9}x$

**Step 1:** Combine under one root: $\sqrt{\frac{8x^3}{27x}} = \sqrt{\frac{8x^2}{27}} = \frac{2\sqrt{2}\,x}{3\sqrt{3}}$.
**Step 2:** Rationalize: $\frac{2\sqrt{2}\,x\cdot\sqrt{3}}{3\cdot 3} = \frac{2\sqrt{6}}{9}x$.

**41** **Answer: A** $2(3x-2)(3x-2)$

**Step 1:** Common factor 2: $2(9x^2 - 12x + 4)$.
**Step 2:** $9x^2 - 12x + 4 = (3x-2)^2$.
**Step 3:** $2(3x-2)(3x-2)$.

**42** **Answer: C** $\frac{17}{13}$

**Step 1:** $\tan x = \frac{5}{12}$: opposite 5, adjacent 12, hypotenuse $\sqrt{25+144} = 13$.
**Step 2:** $\sin x = \frac{5}{13}$, $\cos x = \frac{12}{13}$.
**Step 3:** $\sin x + \cos x = \frac{17}{13}$.

**43** **Answer: C** $x \le -3$

**Step 1:** Subtract 3: $-x \ge 3$.
**Step 2:** Multiply by $-1$ and flip the sign: $x \le -3$.

**44** **Answer: D** $\frac{\sqrt{3}}{2}$

**Step 1:** $\frac{36}{48} = \frac{3}{4}$.
**Step 2:** $\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$.

**45** **Answer: B** 85

**Step 1:** $\overline{CB}$ passes through the center, so it is a diameter; angles inscribed in a semicircle are right: $\angle CAB = \angle CDB = 90^\circ$.
**Step 2:** In $\triangle ABC$: $\angle ACB = 90^\circ - 40^\circ = 50^\circ$.
**Step 3:** In $\triangle BCD$: $\angle CBD = 90^\circ - 55^\circ = 35^\circ$. Sum $= 85^\circ$.

Rabaa Farrag

**46** **Answer: C** 65

**Step 1:** Convert the time to minutes: $3$ h $20$ min $= 200$ min.
**Step 2:** $y = 0.34(200) - 3 = 68 - 3 = 65$.

**47** **Answer: D** $-4x^3 - 6x^2 + 4x$

**Step 1:** Multiply the binomials first: $(2x - 1)(x + 2) = 2x^2 + 4x - x - 2 = 2x^2 + 3x - 2$.
**Step 2:** Multiply by $-2x$: $-4x^3 - 6x^2 + 4x$.

**48** **Answer: B** $\frac{520}{7}$

**Step 1:** Total of the first $3$ marks: $3 \times 80 = 240$. Total of the other $4$: $4 \times 70 = 280$.
**Step 2:** Overall mean $= \frac{240+280}{7} = \frac{520}{7}$.

**49** **Answer: B** $x^2 - 5x + 12$

**Step 1:** $f(x) - g(x) = (x^2 + 3) - (5x - 9)$.
**Step 2:** $= x^2 + 3 - 5x + 9 = x^2 - 5x + 12$.

**50** **Answer: A** 15

**Step 1:** Note: the figure as printed does not give enough information to derive $x$ rigorously (only the two segments $15$ and $25$ and two equal sides are marked).
**Step 2:** The official answer key gives $x = 15$.

Rabaa Farrag